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Super-resolution for reliable industrial inspection under mixed degradations

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ABSTRACT

Industrial defect inspection is sensitive to image degradation because blur, noise, compression, and limited spatial resolution can obscure subtle structural variations, while super-resolution (SR) reconstruction may introduce high-frequency artifacts that interfere with downstream inspection. To address this problem, we propose SynSR, an inspection-oriented SR framework for mixed degradations. Given a degraded low-resolution image, SynSR first performs structural prior extraction, where gradient and Laplacian responses are encoded into a compact prior to characterize local structural variations. The prior is then introduced into the restoration backbone through the Prior-Guided Feature Transfer Layer (PG-FTL), where a Global PriorGate adaptively modulates intermediate feature updates according to the structural prior, the current feature response, and optional degradation information. Before reconstruction, Artifact-Oriented Convolution (AOR-Conv) regulates intermediate responses in a spatial- and channel-aware manner to suppress unstable high-frequency activations that may propagate into the SR output. Different SR front-ends are finally evaluated under a unified SR-to-inspection protocol with matched test degradation conditions and unchanged downstream inspection architectures. Experiments on MVTec AD and MPDD show that SynSR achieves a favorable balance between reconstruction quality and downstream anomaly detection and localization performance under mixed degradations.

1. Introduction

With the rapid advancement of intelligent manufacturing and industrial automation, computer vision-based surface defect inspection has become an important component of industrial quality control and online production monitoring (Wang et al., 2025). However, reliable image acquisition cannot always be guaranteed in practical production environments. Mechanical vibration, illumination variation, specular reflection, sensor noise, and other acquisition disturbances may reduce effective spatial resolution, obscure fine structural details, and weaken local defect cues (Martin et al., 2021). This issue is particularly critical for small, low-contrast, and morphologically irregular defects, which can be easily obscured by complex background textures or degradation-induced artifacts (Bergmann et al., 2019). Although lightweight preprocessing operations such as denoising, illumination correction, and contrast enhancement are commonly applied before inspection, they have limited ability to recover fine spatial information that has already been lost during low-resolution acquisition (Wang et al., 2025).

The problem becomes more challenging because image degradations in real industrial scenarios rarely occur in isolation. Blur, noise, compression, and limited spatial resolution may coexist or interact, producing observations that are more complex than those generated by a single degradation operator (Nahar et al., 2026). Similar challenges have motivated recent research on image restoration under multiple and unknown degradations (Jiang et al., 2025). Under such conditions, genuine structural information may be weakened while degradation-induced high-frequency responses become more prominent. Super-resolution (SR) therefore provides a natural front-end for industrial inspection because it can recover spatial details from low-resolution observations and provide richer visual information for subsequent recognition (Liang et al., 2021). In this study, the mixed-degradation setting serves as a controlled evaluation condition for investigating how SR can provide reliable restoration when multiple degradation factors coexist.

However, a straightforward reconstruct-then-detect pipeline does not necessarily improve downstream inspection performance. Conventional SR methods are mainly optimized for reconstruction fidelity or perceptual image quality, and their objectives are not explicitly aligned with downstream recognition or inspection tasks (Kim et al., 2024). Practical blind SR methods have improved restoration robustness through more realistic degradation modeling and training

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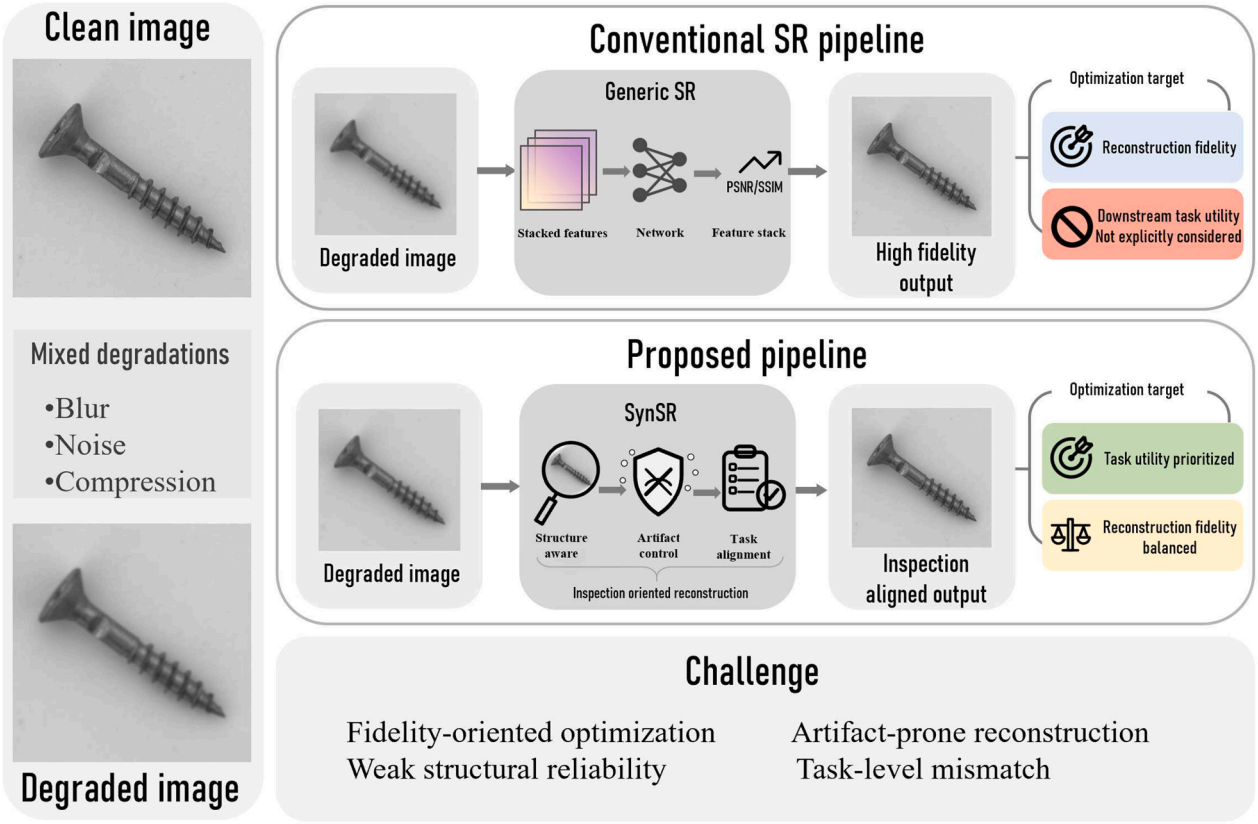


Fig. 1. Conceptual comparison of generic SR and SynSR for industrial defect inspection under mixed degradations.

strategies (Zhang et al., 2021). Nevertheless, reconstruction under complex degradations may still introduce ringing, over-sharpening, or spurious high-frequency textures (Wang et al., 2021). Such responses may make reconstructed images appear visually sharper while altering local texture statistics or boundary structures, which is particularly problematic for industrial anomaly inspection because these reconstruction-induced patterns may be misinterpreted as defect-related cues.

This limitation can be overlooked when SR methods are evaluated primarily from a reconstruction perspective. PSNR and SSIM are widely used to quantify pixel-level fidelity and structural similarity (Huynh-Thu & Ghanbari, 2008; Wang et al., 2004), but improvements in these reconstruction metrics do not necessarily translate into better downstream recognition performance (Kim et al., 2024). A reconstructed image may appear clearer while still altering local boundary statistics or introducing subtle artifacts that affect subsequent predictions (Deo et al., 2025). This distinction is particularly important for industrial inspection, where small anomalies are often identified through subtle local deviations. As illustrated in Fig. 1, the key question is therefore not only whether SR produces a higher-quality image, but also whether the reconstructed image preserves sufficiently reliable structural information without introducing misleading responses for downstream defect detection and localization.

To address this problem, we propose SynSR, an inspection-oriented super-resolution framework for mixed degradations. SynSR first extracts lightweight structural cues from the degraded input and encodes them into a compact prior. The prior is then introduced into the restoration backbone through a Prior-Guided Feature Transfer Layer (PG-FTL), where a Global PriorGate modulates intermediate feature updates according to the structural prior, the current feature response, and optional degradation information. Before reconstruction, Artifact-Oriented Convolution (AOR-Conv) further regulates unstable spatial and channel responses that may otherwise propagate into the SR output. These components are designed to preserve input-supported structural information while reducing reconstruction-induced interference with downstream inspection. We further evaluate SynSR under a unified SR-to-inspection protocol with unchanged downstream inspector architectures and matched test degradation conditions.

The main contributions of this paper are summarized as follows:

- We propose SynSR, an inspection-oriented super-resolution framework for industrial images under mixed degradations. Unlike reconstruction-oriented SR methods, SynSR emphasizes the preservation of informative structural cues while reducing reconstruction-induced responses that may interfere with downstream defect detection and localization.
- We develop a structure-guided restoration strategy that integrates structural prior extraction, Prior-Guided Feature Transfer Layers (PG-FTLs) with a Global PriorGate, and AOR-Conv. The structural prior provides input-supported guidance for intermediate feature transfer, while AOR-Conv regulates unstable spatial and channel responses before reconstruction.
- We establish a unified SR-to-inspection evaluation protocol in which different SR front-ends are assessed under matched test degradation conditions with unchanged downstream inspector architectures. This protocol evaluates SR from both reconstruction and task-level perspectives, enabling the effects of different SR front-ends

on downstream anomaly detection and localization to be examined directly.

2. Related work

2.1. Industrial defect detection and anomaly detection

Industrial visual inspection has evolved from handcrafted feature-based methods to deep learning-based defect detection and anomaly detection. Supervised approaches commonly formulate inspection as object detection or semantic segmentation and can achieve accurate defect recognition when sufficient annotations are available (Deshpande et al., 2024; Ma et al., 2024). Recent studies have further improved local–global feature interaction for industrial defect detection. For example, EDGL-Net introduces a dynamic global–local architecture for real-time metal surface defect detection in industrial edge environments (Gao et al., 2026). However, supervised methods remain dependent on labeled defective samples, which are often scarce in practical manufacturing scenarios.

To reduce this dependence, anomaly detection methods learn representations of normal patterns and identify defects as deviations from normality (Li et al., 2025). Existing approaches include reconstruction-based methods (Zavrtanik et al., 2021), feature-modeling methods (Defard et al., 2021; Zhang et al., 2023), and memory-based matching methods (Roth et al., 2022; Yu et al., 2021). More recently, vision–language contrastive learning has also been explored for surface anomaly detection in consumer-electronics manufacturing (Gao et al., 2024). Despite these advances, most existing methods focus primarily on improving downstream inspection models and generally assume that the input images contain sufficiently reliable local structural information.

2.2. Super-resolution under complex degradations

Single-image super-resolution has progressed from convolutional neural network (CNN)-based architectures to Transformer-based and multi-scale attention models. SwinIR is a representative Transformer-based approach that employs shifted-window attention for image restoration and super-resolution (Liang et al., 2021). Multi-scale and frequency-domain modeling have also been explored. For example, a multi-scale fast Fourier transform based attention network combines multi-scale representation with frequency-domain attention for remote-sensing image super-resolution (Wang et al., 2023). These methods improve feature representation and detail recovery, but are primarily developed from the perspective of reconstruction quality.

Practical and blind SR methods further address complex degradations beyond simple bicubic downsampling. BSRGAN combines blur, resampling, and noise through a shuffled degradation pipeline (Zhang et al., 2021), while Real-ESRGAN introduces higher-order degradation modeling and explicitly considers ringing and overshoot artifacts (Wang et al., 2021). Although these methods improve restoration robustness under complex degradations, reconstructed high-frequency content may still alter local structural statistics or introduce artifacts that are unfavorable for downstream anomaly inspection.

2.3. Task-oriented super-resolution for industrial inspection

Task-oriented super-resolution seeks to align image restoration with downstream recognition rather than optimizing reconstruction fidelity alone. Task-driven SR methods introduce recognition-related objectives to improve the usefulness of reconstructed images for subsequent vision tasks (Haris et al., 2021). SR4IR further shows that improvements in reconstruction metrics do not necessarily translate into better recognition performance (Kim et al., 2024). This distinction is particularly important for industrial inspection, where small defects are often characterized by subtle local structural variations and reconstruction artifacts may be misinterpreted as defect-related cues.

Overall, existing industrial inspection methods mainly improve detector-side representations, whereas practical SR methods primarily focus on reconstruction quality or degradation robustness. The influence of SR reconstruction on industrial anomaly detection and pixel-level localization under mixed degradations remains insufficiently studied. This gap motivates our inspection-oriented SR formulation, which focuses on preserving informative structural cues while reducing reconstruction-induced interference with downstream inspection.

3. Method

3.1. Overview of SynSR

As illustrated in Fig. 2, SynSR follows a shallow–deep–head architecture. Given a degraded input, a shallow feature extraction layer first maps the image into the feature space, while Sobel–Laplacian responses are encoded into a compact structural prior. The deep restoration stage consists of stacked Residual Prior-Gated Blocks (RPGBs), within which Prior-Guided Feature Transfer Layers (PG-FTLs) modulate intermediate feature updates using the structural prior, the current feature response, and optional degradation information when enabled. AOR-Conv subsequently regulates unstable spatial and channel responses before reconstruction. Finally, a PixelShuffle-based reconstruction head generates the super-resolved image for downstream inspection. The following subsections describe structural prior extraction, prior-guided feature transfer, and artifact-oriented response regulation in detail.

3.2. Prior extraction

In practice, subtle defect cues are often weakened by downsampling, blur, and noise, making direct feature restoration unstable if no external structural guidance is provided. For this reason, SynSR introduces a lightweight structural prior extracted directly from the degraded input, which serves as a global conditioning signal for subsequent feature transfer.

The structural prior is not intended to identify defect regions semantically. Instead, it provides low-level guidance about local structural variations in the degraded observation. Specifically, given the degraded input image $\mathbf{y} \in \mathbb{R}^{H_y \times W_y \times 3}$, where $(H_y, W_y) = (H/s, W/s)$, we construct a two-channel prior map by combining the Sobel gradient magnitude and the absolute Laplacian response:

$$\mathbf{P}_{\text{map}} = \text{cat}\big(\mathcal{G}(\mathbf{y}),\ \mathcal{L}(\mathbf{y})\big) \in \mathbb{R}^{2 \times H_y \times W_y}, \tag{1}$$

where $\mathcal{G}(\cdot)$ denotes the Sobel gradient magnitude and $\mathcal{L}(\cdot)$ denotes the absolute Laplacian response. The Sobel term describes first-order local intensity transitions, whereas the Laplacian term provides complementary second-order information about local structural variation. These responses are used as auxiliary structural guidance for restoration rather than as semantic indicators of defect locations.

To make this prior suitable for global conditioning, we further compress the spatial prior map into a compact embedding through a lightweight prior encoder $\text{PE}(\cdot)$:

$$\mathbf{P} = \text{PE}\big(\mathbf{P}_{\text{map}}\big) \in \mathbb{R}^{C_p}. \tag{2}$$

The resulting vector $\mathbf{P}$ is shared across PG-FTLs and used to guide feature modulation throughout the restoration backbone. In this way, the extracted prior acts as a stable structural cue under mixed degradations and guides subsequent feature transfer toward input-supported structural responses rather than blindly enhancing all high-frequency patterns.

3.3. Prior-guided feature transfer

The role of prior-guided feature transfer in SynSR is not to add an auxiliary branch to the backbone, but to directly regulate how intermediate features are updated during restoration. Under mixed degradations, attention-based feature propagation may still amplify unstable or

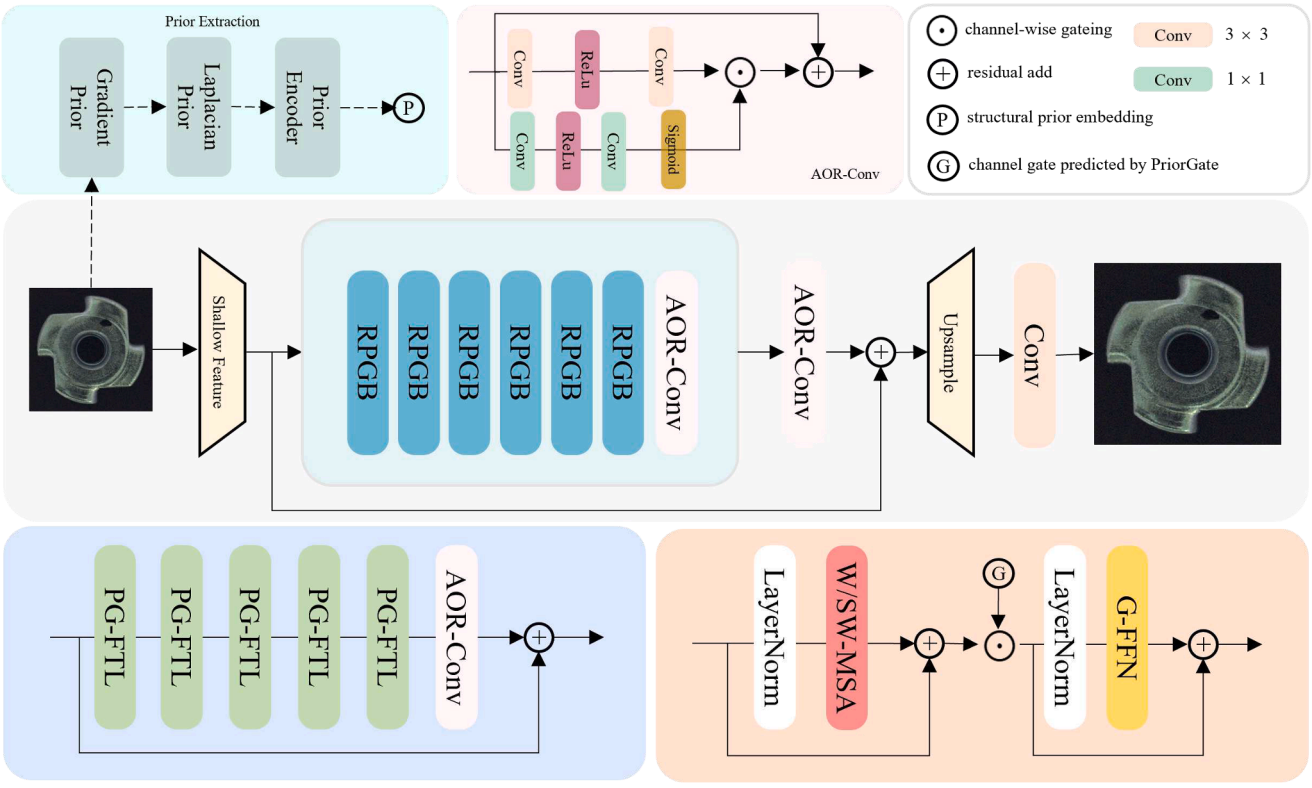


Fig. 2. Overview of SynSR. SynSR adopts a shallow–deep–head structure with residual prior-gated blocks and a PixelShuffle upsampling head. A global PriorGate modulates the attention update, conditioned on a structural prior $\mathbf{P}$ and an optional degradation embedding $\mathbf{d}_{\text{embed}}$. AOR-Conv regulates intermediate responses to suppress artifacts.

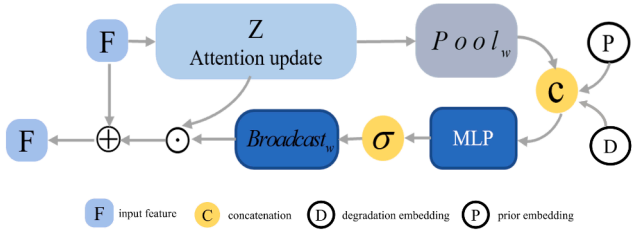


Fig. 3. Global PriorGate used in PG-FTL.

artifact-prone responses if it is driven only by the degraded input itself. This is particularly problematic for industrial inspection, where weak boundary evidence and subtle local structures are often more important than globally plausible texture recovery. To address this issue, SynSR injects the extracted structural prior into the feature update process, so that restoration is biased toward structurally informative responses rather than unconstrained high-frequency enhancement.

As shown in Fig. 3, each PG-FTL takes token features $\mathbf{F} \in \mathbb{R}^{N\times C}$ as input and produces an attention update $\mathbf{Z} \in \mathbb{R}^{N\times C}$ through window-based or shifted-window self-attention. Instead of directly adding $\mathbf{Z}$ back to $\mathbf{F}$, we first summarize the update by window/token pooling:

$$\mathbf{z}_w = \text{Pool}_w(\mathbf{Z}) \in \mathbb{R}^C. \tag{3}$$

This pooled vector captures the global response tendency of the current attention update and provides a compact context descriptor for subsequent conditioning.

We then combine the pooled feature context with the structural prior and the optional degradation embedding. Let

$$\mathbf{p} = \text{cat}\left(\mathbf{P}, \mathbf{d}_{\text{embed}}\right) \in \mathbb{R}^{C_p+C_e}, \tag{4}$$

where $\mathbf{P}$ is the extracted structural prior and $\mathbf{d}_{\text{embed}}$ is set to zero when degradation conditioning is disabled. The conditioning vector is formed as

$$\tilde{\mathbf{p}} = \text{cat}\left(\mathbf{p}, \mathbf{z}_w\right) \in \mathbb{R}^{C_p+C_e+C}. \tag{5}$$

A lightweight multilayer perceptron (MLP) then predicts a channel-wise modulation gate:

$$\mathbf{g} = \sigma\big(\text{MLP}(\tilde{\mathbf{p}})\big) \in (0,1)^C, \tag{6}$$

where $\sigma(\cdot)$ denotes the sigmoid function. The gate is broadcast along the token dimension and used to modulate the attention update before residual fusion:

$$\mathbf{F}' = \mathbf{F} + \big(\mathbf{Z} \odot \mathbf{g}\big). \tag{7}$$

This formulation makes feature transfer explicitly conditioned on both the current feature response and the extracted structural prior. As a result, the network can enhance structurally supported responses while reducing the indiscriminate amplification of unstable high-frequency patterns. Rather than relying solely on the current feature response, PG-FTL uses the low-level structural prior as auxiliary guidance to modulate intermediate feature transfer. Importantly, the prior itself does not distinguish anomalous structures from normal textures; instead, it provides input-supported structural information that is incorporated into the learned restoration process.

3.4. Artifact-oriented convolution

Prior-guided feature transfer helps SynSR determine which structural responses should be emphasized, but this alone is insufficient for reliable industrial inspection. In practice, super-resolution reconstruction may still introduce ringing artifacts and spurious textures, especially under mixed degradations and weak structural evidence. Such responses are problematic for downstream inspection because they can distort local

statistics and be mistakenly amplified as defect evidence. For this reason, SynSR further introduces an artifact-oriented regulator to constrain intermediate restoration behavior before image reconstruction.

Specifically, we use AOR-Conv as a lightweight dual-branch regulation module. Given an intermediate feature map $\mathbf{U} \in \mathbb{R}^{C\times H\times W}$, the content branch predicts a main response

$$\mathbf{Y}_{\text{main}} = \text{Conv}_{3\times3}\Big(\text{ReLU}\big(\text{Conv}_{3\times3}(\mathbf{U})\big)\Big), \tag{8}$$

while the modulation branch generates a spatial-channel gate

$$\mathbf{g}_{\text{aor}} = \sigma\Big(\text{Conv}_{1\times1}\big(\text{ReLU}(\text{Conv}_{1\times1}(\mathbf{U}))\big)\Big) \in (0,1)^{C\times H\times W}, \tag{9}$$

where $\sigma(\cdot)$ denotes the sigmoid function.

The final regulated output is obtained by element-wise gating:

$$C_{\text{AOR}}(\mathbf{U}) = \mathbf{Y}_{\text{main}} \odot \mathbf{g}_{\text{aor}}, \tag{10}$$

This design uses the learned gate to adaptively reweight intermediate responses in a spatial- and channel-aware manner. Since the gate values lie in $(0,1)$, artifact-prone activations can be suppressed while structurally useful responses are preserved, providing a lightweight form of artifact control before reconstruction.

Within SynSR, AOR-Conv is inserted after stacked PG-FTLs inside each RPGB:

$$\text{RPGB}(\mathbf{F},\tilde{\mathbf{p}}) = \mathbf{F} + C_{\text{AOR}}\Big(\Phi(\mathbf{F},\tilde{\mathbf{p}})\Big), \tag{11}$$

where $\Phi(\cdot)$ denotes the composition of PG-FTLs in the current block. Under this design, prior-guided feature transfer and AOR-Conv play complementary roles: the former steers restoration toward input-supported structural responses, while the latter suppresses unstable responses before they propagate to the reconstruction head.

3.5. Matched degradation and unified SR-to-inspection protocol

To evaluate SynSR under consistent mixed-degradation conditions, all compared methods are tested using the same degradation formulation and downsampling factor. The degradation pipeline combines blur, noise, compression, and resolution loss to provide a controlled setting for fair comparison among different SR front-ends. This setting also enables us to examine whether the structure-guided, degradation-aware, and artifact-regulated design of SynSR remains effective as the degradation conditions become increasingly complex.

Matched mixed degradations. Given a high-resolution (HR) normal image $\mathbf{x} \in \mathbb{R}^{H\times W\times3}$ normalized to $[0,1]$, we synthesize its low-resolution (LR) observation through an industrial degradation operator:

$$\mathbf{y} = \mathcal{D}(\mathbf{x}; s, \boldsymbol{\eta}) = \text{Down}_s^{\text{bic}}\Big(\mathcal{J}_q^{\delta_q}\big(\mathcal{N}_{\sigma_n}^{\delta_n}(\mathcal{B}_{\sigma_b,k}(\mathbf{x}))\big)\Big), \tag{12}$$

where $s \in \{2,4\}$ denotes the downsampling factor, and $\boldsymbol{\eta} = \{\sigma_b, k, \sigma_n, q, \delta_n, \delta_q\}$ collects the degradation strengths and activation switches. Here, $\mathcal{B}$, $\mathcal{N}$, and $\mathcal{J}$ denote Gaussian blur, additive Gaussian noise, and JPEG compression, respectively. For blur, the kernel size is set to $k \in \{7,9\}$ according to severity. For noise and JPEG compression, the activation switches are sampled as $\delta_n, \delta_q \sim \text{Bernoulli}(p)$. Unless otherwise stated, all compared methods use the same degradation operator $\mathcal{D}(\cdot)$ and the same downsampling factor s.

Unified SR→inspection protocol. Given the degraded input $\mathbf{y}$, SynSR reconstructs an HR image $\hat{\mathbf{x}}$ as

$$\hat{\mathbf{x}} = f_\theta(\mathbf{y}; \mathbf{d}), \tag{13}$$

where $\mathbf{d}$ is an optional degradation descriptor used only when degradation conditioning is enabled. For task-level evaluation, SR is treated as the front-end of the inspection pipeline: downstream inspectors are trained using only normal samples in the reconstructed domain and are evaluated on reconstructed test images. Keeping the downstream architecture, data split, and evaluation procedure unchanged reduces detector-side confounding and facilitates comparison of how different SR front-ends affect downstream performance.

This protocol serves two purposes: it enables comparison between SynSR and competing SR baselines under common test conditions, and it assesses whether SR reconstruction benefits downstream defect detection and localization beyond reconstruction fidelity alone. Overall, SynSR addresses two main challenges under mixed degradations: the loss of useful structural information and the emergence of unstable high-frequency responses during reconstruction. Structural prior extraction and prior-guided feature transfer provide auxiliary guidance for recovering local structures, while AOR-Conv regulates unstable responses before reconstruction. Importantly, the Sobel–Laplacian prior provides low-level structural cues rather than semantic defect awareness. Strong normal textures or residual noise may therefore also produce strong prior responses, whereas smooth or low-contrast anomalies may yield relatively weak responses. This limitation is further examined through defect-region reconstruction analysis in the experimental section.

4. Experiments

4.1. Experimental objectives

The experiments evaluate SynSR from three complementary perspectives: reconstruction robustness under standard and mixed degradations, the influence of different SR front-ends on downstream anomaly detection and localization, and the contribution of the proposed components through controlled ablation studies. All downstream comparisons use fixed inspector architectures and data splits to reduce detector-side confounding.

4.2. Data preparation and degradation settings

We conduct experiments on MVTec AD (Bergmann et al., 2019) and MPDD (Jezek et al., 2021). For SR evaluation, the original high-resolution (HR) images are used as references and degraded to generate the corresponding low-resolution (LR) observations. For downstream anomaly inspection, we follow the standard one-class setting: only normal training samples are used to train the inspectors, while the official test sets containing both normal and anomalous samples are used for evaluation.

Following the degradation formulation described in Section 3.5, we consider standard bicubic downsampling and the synthetic mixed-degradation setting for $s \in 2,4$. Gaussian noise and JPEG compression are independently activated with probability $p = .7$. Three fixed test severities, Mild (L), Moderate (M), and Severe (H), are used for reproducible evaluation, with their parameter ranges summarized in Table 1

4.3. Comparison methods and implementation details

For reconstruction comparison, we include representative CNN-based, Transformer-based, and practical SR methods. EDSR (Lim et al., 2017) and RCAN (Zhang et al., 2018) represent convolutional SR architectures, while SwinIR (Liang et al., 2021) and HAT (Chen et al., 2023) represent Transformer-based SR architectures. Under mixed degradations, Real-ESRGAN (Wang et al., 2021) and StableSR (Wang et al., 2024a) are further included as representative practical restoration methods. For all competing methods, we use their officially released pre-trained models without additional task-specific retraining or adaptation. All methods are evaluated using the same test images, degraded inputs, scale factors, and evaluation metrics. This protocol is intended to assess the practical robustness of different SR front-ends when directly applied to industrial images under common evaluation conditions.

SynSR is implemented in PyTorch 2.0.1 with CUDA 11.7 using a KAIR-style training framework and is trained on a single NVIDIA GeForce RTX 3090 GPU with 24 GB of memory. We use the Adam optimizer with $\beta_1 = 0.9$, $\beta_2 = 0.999$, and the weight decay set to zero. The initial learning rate is set to 5×10^{-5}, and the batch size is 8. For $\times2$ SR, the HR/LR training patch sizes are 128×128 and 64×64. The learning

Table 1
Parameter values of degradation settings.

Setting	Blur (k, σ_b)	Noise σ_n	JPEG q
×2			
×2-L (Mild)	(7, [0.3, 0.7])	[1, 4]/255	[75, 95]
×2-M (Moderate)	(7, [0.7, 1.2])	[3, 8]/255	[50, 85]
×2-H (Severe)	(9, [1.2, 1.6])	[8, 12]/255	[30, 60]
×4			
×4-L (Mild)	(7, [0.2, 0.5])	[1, 3]/255	[80, 95]
×4-M (Moderate)	(7, [0.5, 0.9])	[2, 6]/255	[60, 90]
×4-H (Severe)	(9, [0.9, 1.2])	[6, 10]/255	[40, 70]

rate is reduced by a factor of 0.5 at 5k, 8k, and 11k iterations using MultiStepLR. For ×4 SR, the HR/LR training patch sizes are 256×256 and 64×64.

The reconstruction objective is based on the Charbonnier loss with a loss weight of 1.0 and $\epsilon = 10^{-9}$. For computational complexity analysis, FLOPs are calculated using a fixed HR output resolution of 1024×1024. Accordingly, the LR input resolution is 512×512 for ×2 SR and 256×256 for ×4 SR.

For downstream inspection, PatchCore and DeSTSeg are used as complementary anomaly inspectors. For each SR front-end, both the normal training images and the test images are reconstructed using the corresponding SR model before being provided to the inspector. The downstream architecture, data split, and evaluation procedure are kept unchanged across different SR front-ends so that the observed performance differences primarily reflect the influence of the reconstructed inputs.

4.4. Evaluation metrics

We evaluate the SR front-end from reconstruction, perceptual, structural, and downstream inspection perspectives. PSNR and SSIM are used to measure pixel-level fidelity and structural similarity with respect to the HR reference, respectively, while LPIPS measures perceptual discrepancy. Higher PSNR and SSIM and lower LPIPS indicate better reconstruction performance.

To quantify gradient-domain distortion, we additionally report GradErr. Let $\hat{\mathbf{x}}$ and $\mathbf{x}$ denote the reconstructed image and the corresponding HR reference, respectively. Before computing GradErr, both images are converted to grayscale and represented on the $[0, 255]$ intensity scale, denoted by $\tilde{\hat{\mathbf{x}}}$ and $\tilde{\mathbf{x}}$. We define the directional Sobel error as

$$\Delta_d(p) = S_d(\tilde{\hat{\mathbf{x}}})(p) - S_d(\tilde{\mathbf{x}})(p), \quad d \in \{x, y\}, \tag{14}$$

and compute

$$\text{GradErr} = \frac{1}{HW} \sum_p \sqrt{\Delta_x(p)^2 + \Delta_y(p)^2}. \tag{15}$$

Here, $S_x(\cdot)$ and $S_y(\cdot)$ denote the 3×3 horizontal and vertical Sobel operators, respectively. A lower GradErr indicates smaller gradient-domain distortion.

For downstream inspection, image-level AUROC (I-AUROC) measures anomaly detection performance, while pixel-level AUROC (P-AUROC) evaluates pixel-wise anomaly localization. We additionally report pixel-level average precision (P-AP), which summarizes the pixel-wise precision–recall relationship and is particularly informative under the strong class imbalance between normal and anomalous pixels. The per-region overlap (PRO) score is used to assess region-level localization by quantifying the overlap between predicted anomalous regions and ground-truth defect regions. Higher I-AUROC, P-AUROC, P-AP, and PRO indicate better downstream inspection performance. For each metric, the dataset-level result is obtained by first computing the category-level scores and then averaging them across all categories.

4.5. Super-resolution evaluation

This subsection evaluates SynSR from the reconstruction perspective under both standard bicubic downsampling and the production-inspired synthetic mixed-degradation setting. We compare SynSR with representative SR baselines on MVTec AD and MPDD at both ×2 and ×4. As shown in Table 2, the compared methods generally achieve strong reconstruction performance under standard bicubic downsampling. The qualitative results in Fig. 4 further show that most methods recover the main object contours and local appearances reasonably well, with relatively small visual differences among the reconstructed images. These results indicate that standard bicubic downsampling provides limited discrimination among different SR methods in this evaluation.

We further evaluate the compared methods under the controlled mixed-degradation setting described in Table 1. As illustrated in Fig. 5, different SR methods produce noticeably different local edge and texture responses under Moderate degradation at ×4, even when their overall reconstructed appearances remain visually plausible. Table 3 shows that reconstruction performance generally deteriorates as the degradation severity increases. SynSR maintains strong PSNR and SSIM performance, particularly at ×4, and achieves low GradErr in most settings, although it is not uniformly optimal across all metrics. For example, Real-ESRGAN obtains lower LPIPS in several MPDD settings, while individual baselines achieve lower GradErr under some conditions. The gradient-domain comparison in Fig. 6 further illustrates that visually plausible reconstructions can still exhibit noticeable local gradient discrepancies.

Overall, these results show that different SR methods exhibit distinct trade-offs across reconstruction fidelity, perceptual quality, and structural consistency. Reconstruction performance alone is therefore insufficient to determine whether an SR front-end is suitable for industrial inspection, motivating the downstream task-level evaluation that follows.

Beyond global reconstruction metrics, we further examine reconstruction quality within annotated defect regions to assess whether small and low-contrast anomalies remain adequately preserved under mixed degradations.

Let $M(p) \in \{0, 1\}$ denote the ground-truth defect mask, $\hat{\mathbf{x}}$ the reconstructed image, and $\mathbf{x}$ the HR reference. We define the defect-region mean absolute error as

$$\text{Defect-MAE} = \frac{\sum_p M(p) \, \|\hat{\mathbf{x}}(p) - \mathbf{x}(p)\|_1}{3 \sum_p M(p)}, \tag{16}$$

where RGB intensities are normalized to $[0, 1]$. We further quantify structural distortion within defect regions using

$$\text{Defect-GradErr} = \frac{1}{6 \sum_p M(p)} \sum_{d \in \{x,y\}} \sum_p M(p) \times \left\| S_d(\hat{\mathbf{x}})(p) - S_d(\mathbf{x})(p) \right\|_1, \tag{17}$$

where $S_x(\cdot)$ and $S_y(\cdot)$ denote the 3×3 horizontal and vertical Sobel operators, respectively, and the error is averaged over the three RGB channels. Lower values of Defect-MAE and Defect-GradErr indicate better preservation of local appearance and structural variations within annotated defect regions.

As shown in Table 4, SynSR achieves the lowest defect-region errors on both datasets under the Moderate mixed-degradation setting at ×4. Compared with bicubic upsampling, SynSR reduces Defect-MAE and Defect-GradErr by 6.7% and 4.7%, respectively, on MVTec AD, and by 6.4% and 4.3%, respectively, on MPDD. SynSR also outperforms SwinIR and StableSR on both defect-region metrics, suggesting that visually sharp reconstruction does not necessarily correspond to faithful preservation of defect-region appearance and local structure. A paired bootstrap analysis with 10,000 image-level resamples provides further statistical support for the improvements over Bicubic and SwinIR, with the 95% confidence intervals for all corresponding error reductions remaining above zero.

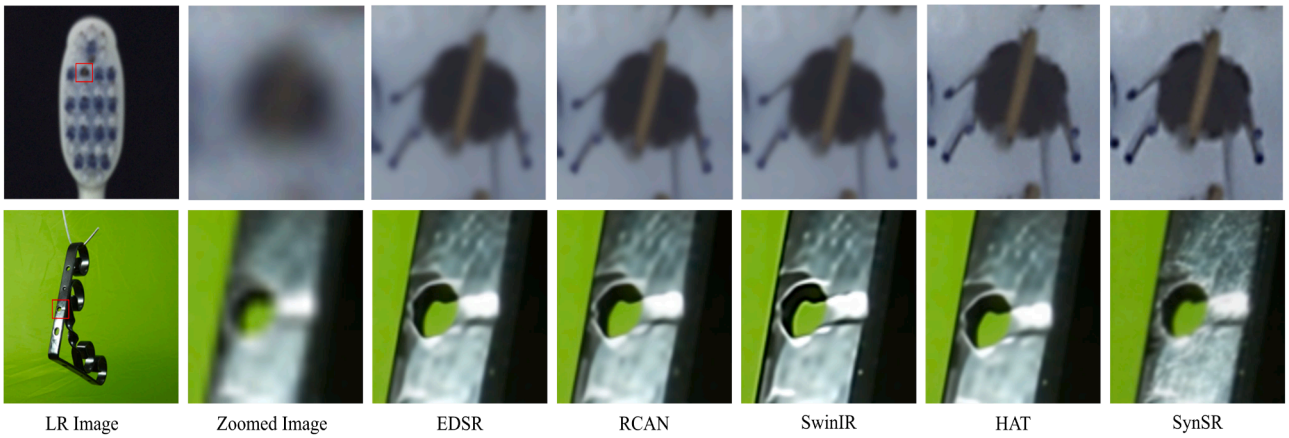


Fig. 4. Qualitative comparison under standard bicubic downsampling.

Table 2
SR comparison under the standard bicubic downsampling setting on MVTec and MPDD.

Method	Scale	Params (M)	FLOPs (G)	MVTec		MPDD	
				PSNR	SSIM	PSNR	SSIM
EDSR (Lim et al., 2017)	×2	40.73	10677.75	41.93	0.9458	42.89	0.9615
RCAN (Zhang et al., 2018)	×2	15.44	4016.04	43.59	0.9758	42.86	0.9617
SwinIR (Liang et al., 2021)	×2	11.72	821.25	43.65	0.9761	43.11	0.9624
HAT (Chen et al., 2023)	×2	20.62	26041.08	43.73	0.9765	44.82	0.9691
SynSR	×2	13.38	823.72	42.89	0.9740	44.91	0.9701
EDSR (Lim et al., 2017)	×4	43.09	3293.35	37.40	0.9139	38.59	0.9206
RCAN (Zhang et al., 2018)	×4	15.59	1044.02	38.43	0.9243	38.68	0.9208
SwinIR (Liang et al., 2021)	×4	11.90	861.26	38.64	0.9267	38.99	0.9219
HAT (Chen et al., 2023)	×4	20.77	26681.29	39.06	0.9310	40.87	0.9353
SynSR	×4	13.69	867.31	39.11	0.9421	40.82	0.9291

Table 3
SR comparison on MVTec AD and MPDD under mixed degradations at three severity levels (L/M/H). Competing methods are evaluated using their officially released pretrained models without additional adaptation, and all methods are tested on the same degraded inputs for each scale factor and severity level.

Dataset	Method	Scale	L				M				H			
			PSNR↑	SSIM↑	LPIPS↓	GradErr↓	PSNR↑	SSIM↑	LPIPS↓	GradErr↓	PSNR↑	SSIM↑	LPIPS↓	GradErr↓
MVTec	EDSR (Lim et al., 2017)	×2	37.11	0.9131	0.1462	16.1297	35.01	0.8621	0.2631	21.1687	31.97	0.7613	0.3679	30.7562
	RCAN (Zhang et al., 2018)	×2	37.09	0.9018	0.1447	15.9876	34.89	0.8556	0.2732	21.2375	31.65	0.7518	0.3842	30.8541
	SwinIR (Liang et al., 2021)	×2	37.16	0.9128	0.1443	15.8561	34.75	0.8465	0.2561	21.0681	31.66	0.7492	0.3866	30.8682
	HAT (Chen et al., 2023)	×2	**40.98**	**0.9524**	0.1400	15.8202	**38.09**	**0.9072**	0.2588	21.1703	**34.40**	**0.8228**	0.4034	31.0362
	Real-ESRGAN (Wang et al., 2021)	×2	32.66	0.8656	0.1822	25.2493	31.96	0.8466	**0.2031**	27.0796	30.58	0.8093	**0.2475**	31.8925
	StableSR (Wang et al., 2024a)	×2	27.75	0.7804	0.3048	42.2801	28.32	0.7842	0.3162	40.3424	28.77	0.7797	0.3321	37.7195
	SynSR	×2	38.69	0.9216	**0.1294**	**13.5674**	36.83	0.8764	0.2417	**20.8647**	33.21	0.7964	0.3217	**27.2597**
	EDSR (Lim et al., 2017)	×4	32.21	0.8457	0.3527	29.7312	31.48	0.8106	0.3626	30.6348	29.62	0.7121	0.4688	32.4387
	RCAN (Zhang et al., 2018)	×4	32.07	0.8489	0.3217	29.6521	31.47	0.8128	0.3612	27.4829	30.07	0.7654	0.4764	31.2347
	SwinIR (Liang et al., 2021)	×4	32.18	0.8305	0.3090	29.3536	31.25	0.8074	0.3730	28.3070	29.98	0.7456	0.4810	30.8317
	HAT (Chen et al., 2023)	×4	33.54	0.8746	0.2968	29.2492	33.78	0.8646	0.3611	28.3533	32.65	0.8242	0.4707	31.0290
	Real-ESRGAN (Wang et al., 2021)	×4	29.53	0.7877	0.2486	29.1246	29.02	0.7715	**0.2610**	28.2547	28.49	0.7555	**0.2825**	29.2492
	StableSR (Wang et al., 2024a)	×4	28.53	0.7923	0.3098	37.6345	28.73	0.7867	0.3223	36.7120	28.90	0.7769	0.3399	35.6077
	SynSR	×4	**34.77**	**0.8877**	**0.2232**	**19.7650**	**34.39**	**0.8764**	0.2617	**20.3258**	**32.81**	**0.8444**	0.3609	**22.6310**
MPDD	EDSR (Lim et al., 2017)	×2	36.41	0.8789	0.1762	17.4895	33.79	0.7842	0.2769	21.9621	30.93	0.7213	0.4037	31.2684
	RCAN (Zhang et al., 2018)	×2	36.32	0.8698	0.1633	17.8421	33.68	0.7624	0.2743	22.4732	30.87	0.7046	0.4176	30.2447
	SwinIR (Liang et al., 2021)	×2	36.34	0.8781	0.1409	17.3713	33.76	0.7880	0.2853	22.4945	30.91	0.7029	0.4151	30.2053
	HAT (Chen et al., 2023)	×2	**38.91**	**0.9211**	0.1319	17.3538	36.70	0.8645	0.2760	22.6079	33.47	0.7876	0.4165	30.4461
	Real-ESRGAN (Wang et al., 2021)	×2	35.09	0.8933	**0.0942**	**15.7669**	34.67	0.8829	**0.1045**	**16.5984**	34.01	**0.8739**	**0.1166**	**17.2480**
	StableSR (Wang et al., 2024a)	×2	31.50	0.8603	0.2538	20.1375	32.08	0.8520	0.2799	19.4884	32.16	0.8366	0.3039	19.5282
	SynSR	×2	37.84	0.9047	0.1284	16.8753	**36.89**	**0.8997**	0.1469	17.2175	**34.27**	0.7954	0.3157	18.2175
	EDSR (Lim et al., 2017)	×4	29.76	0.8314	0.2907	19.2158	30.39	0.7987	0.3607	**17.0799**	29.37	0.7231	0.4614	28.1137
	RCAN (Zhang et al., 2018)	×4	29.68	0.8311	0.2872	26.0183	30.14	0.7987	0.3591	24.3893	29.21	0.7218	0.4587	28.0321
	SwinIR (Liang et al., 2021)	×4	29.49	0.8282	0.2815	25.9894	30.06	0.7949	0.3509	24.2777	29.12	0.7193	0.4599	27.9934
	HAT (Chen et al., 2023)	×4	31.56	0.8728	0.2862	25.5403	32.36	0.8580	0.3679	24.7968	31.99	0.8122	0.4897	28.3048
	Real-ESRGAN (Wang et al., 2021)	×4	32.83	0.8641	**0.1554**	19.1762	31.48	0.8434	**0.1321**	21.9191	31.20	0.8289	**0.1442**	23.7272
	StableSR (Wang et al., 2024a)	×4	31.57	0.8614	0.2774	19.4074	32.18	0.8569	0.2945	18.7057	32.15	0.8427	0.3116	**18.9833**
	SynSR	×4	**33.29**	**0.8866**	0.2491	**17.0104**	**33.67**	**0.8801**	0.2578	17.1336	**32.58**	**0.8519**	0.3554	18.9904

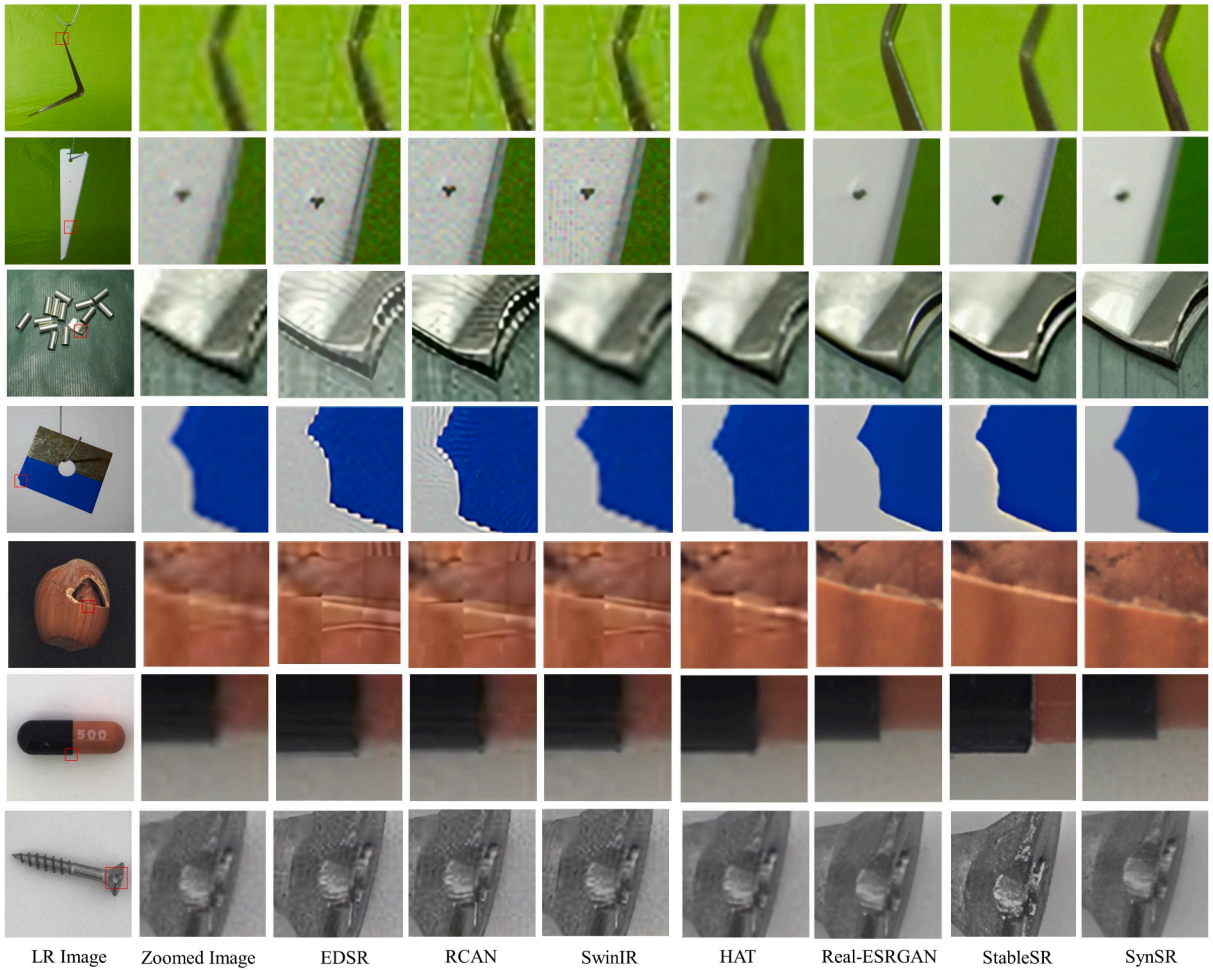


Fig. 5. Qualitative comparison of representative SR methods under the mixed-degradation setting (Moderate severity, ×4).

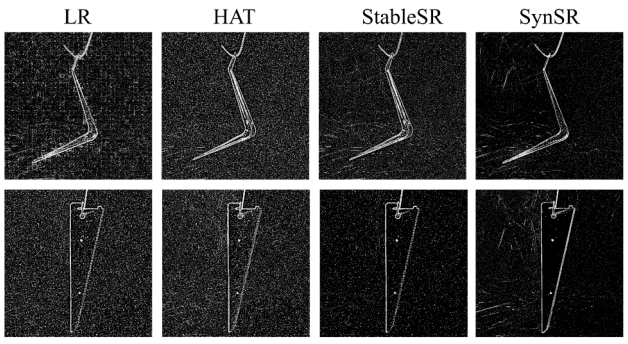


Fig. 6. Gradient-domain comparison under the mixed-degradation setting.

Table 4
Defect-region reconstruction under Moderate mixed degradation at ×4. Values are scaled by 10^2.

Dataset	Method	N	Defect-MAE↓	Defect-GradErr↓
MVTec AD	Bicubic	219	2.306	8.425
	SwinIR	219	2.937	11.105
	StableSR	219	4.573	15.025
	SynSR	219	**2.150**	**8.025**
MPDD	Bicubic	33	4.273	16.278
	SwinIR	33	5.432	21.598
	StableSR	33	4.941	18.732
	SynSR	33	**3.999**	**15.586**

4.6. Downstream defect inspection evaluation

Following the unified SR-to-inspection protocol described above, we evaluate different SR front-ends using PatchCore and DeSTSeg under the Moderate mixed-degradation setting at ×4. In addition to the learned SR methods, the degraded input, bicubic upsampling, and original HR images are included as reference conditions. For each condition, the inspector is trained on normal samples prepared under the corresponding input setting and evaluated on test samples prepared consistently under the same setting, while the downstream architecture, data split, training procedure, and evaluation protocol are kept unchanged.

As shown in Table 5, image-level anomaly detection and pixel-level localization exhibit different sensitivities to the SR front-end. PatchCore I-AUROC remains high for most methods, and the best-performing method varies across datasets. In contrast, more pronounced differences emerge in localization-oriented metrics. SynSR achieves the highest DeSTSeg P-AUROC and PRO among the evaluated learned SR front-ends on both MVTec AD and MPDD. The reference conditions further show that bicubic upsampling improves DeSTSeg localization over the degraded input on both datasets, while the original HR images serve as a clean-image reference. SynSR further improves PRO over bicubic upsampling on both datasets.

The qualitative results in Fig. 7 further illustrate that visually sharper reconstruction does not necessarily lead to better anomaly localization. StableSR can produce detailed reconstructions in which anomalous regions remain visually recognizable. Quantitatively, Table 5 shows that SynSR achieves higher DeSTSeg P-AUROC and PRO than StableSR on

Table 5
Downstream inspection results under the Moderate mixed-degradation setting at ×4 on MVTec AD and MPDD. Degraded input, bicubic upsampling, and original HR images are included as reference conditions. Best and second-best values are highlighted only among learned SR front-ends.

Dataset	Input / Method	PatchCore		DeSTSeg		
		I-AUROC↑	P-AUROC↑	P-AUROC↑	PRO↑	I-AUROC↑
MVTec AD	Degraded input	0.9957	0.9885	0.9762	0.8880	0.9817
	Bicubic	0.9967	0.9884	0.9812	0.9084	0.9909
	Original HR	0.9861	0.9878	0.9879	0.9617	0.9818
	EDSR (Lim et al., 2017)	0.9825	0.9437	0.9768	0.8941	0.9814
	RCAN (Zhang et al., 2018)	0.9813	0.9457	0.9770	0.8947	0.9805
	SwinIR (Liang et al., 2021)	0.9834	0.9793	0.9774	0.8961	0.9811
	HAT (Chen et al., 2023)	0.9865	0.9869	0.9777	0.8974	0.9807
	Real-ESRGAN (Wang et al., 2021)	0.9832	0.9539	0.9345	0.7733	**0.9941**
	StableSR (Wang et al., 2024a)	**0.9989**	0.9862	0.9757	0.9263	0.9814
	SynSR (Ours)	0.9934	**0.9873**	**0.9813**	**0.9411**	0.9863
MPDD	Degraded input	0.9846	0.9964	0.9564	0.8739	0.8101
	Bicubic	0.9829	0.9964	0.9673	0.9102	0.8714
	Original HR	0.9770	0.9968	0.9745	0.9240	0.8715
	EDSR (Lim et al., 2017)	0.9816	0.9954	0.9612	0.8899	0.8348
	RCAN (Zhang et al., 2018)	0.9798	0.9959	0.9749	0.9242	**0.8756**
	SwinIR (Liang et al., 2021)	0.9824	0.9965	0.9614	0.8891	0.8265
	HAT (Chen et al., 2023)	0.9854	0.9966	0.9614	0.8897	0.8266
	Real-ESRGAN (Wang et al., 2021)	0.9877	0.9952	0.9406	0.8547	0.7935
	StableSR (Wang et al., 2024a)	0.9877	0.9967	0.9534	0.8742	0.7754
	SynSR (Ours)	**0.9947**	**0.9974**	**0.9887**	**0.9719**	0.8198

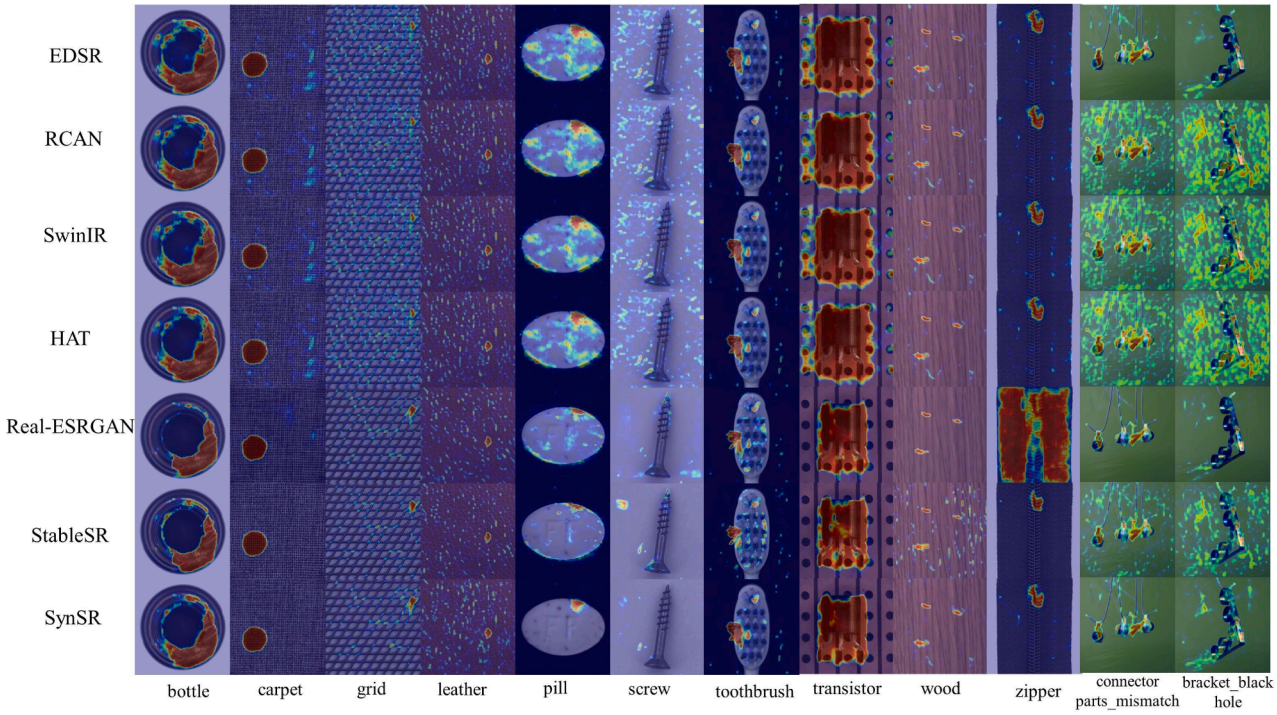


Fig. 7. Qualitative comparison of anomaly heatmaps on representative MVTec categories under the Moderate mixed-degradation setting at ×4.

both datasets. This suggests that downstream localization depends not only on visual sharpness but also on how reconstruction affects local texture and boundary characteristics. Because pixel-level localization is particularly sensitive to such local changes, perceptually plausible high-frequency modifications may still affect the local responses used for spatial anomaly localization. These results indicate that the suitability of an SR front-end for industrial inspection should be assessed using downstream localization performance in addition to visual reconstruction quality.

To further examine localization robustness across degradation severities, we additionally evaluate representative SR front-ends using DeSTSeg on MVTec AD under Mild and Severe mixed degradations at ×4, complementing the Moderate-degradation results reported in Table 5. We report P-AUROC and PRO to assess pixel-level and region-level localization performance under different degradation severities.

As shown in Table 6, the performance differences among the compared methods are relatively small under Mild degradation, where much of the structural information is still preserved. As the degradation severity increases, the differences become more pronounced. Under Severe degradation, SynSR achieves a P-AUROC of 0.9688 and a PRO of 0.9121, outperforming Bicubic and slightly exceeding StableSR on both metrics. Together with the Moderate-degradation results in Table 5, these re-

Table 6
Downstream localization performance of DeSTSeg on MVTec AD under Mild and Severe mixed degradations at ×4.

	Mild		Severe	
Method	P-AUROC	PRO	P-AUROC	PRO
Bicubic	**0.9871**	0.9455	0.9560	0.8290
StableSR	0.9829	0.9484	0.9676	0.9030
SynSR	0.9843	**0.9511**	**0.9688**	**0.9121**

Table 7
Ablation with component toggles and downstream metrics.

Variant	Prior	AOR	Deg-cond	P-AP↑	PRO↑	GradErr↓
Backbone only				0.5326	0.8824	30.2945
+ Prior	✓			0.5842	0.8923	28.3642
Prior + AOR	✓	✓		0.6127	0.8862	25.7562
Prior + Deg-cond	✓		✓	0.6358	0.9237	23.1289
Full (Ours)	✓	✓	✓	0.6659	0.9411	20.3258

Table 8
Known-d and blind degradation conditioning on MVTec AD under Moderate mixed degradations at ×4.

Variant	PatchCore		DeSTSeg		
	I-AUROC↑	P-AUROC↑	P-AUROC↑	PRO↑	I-AUROC↑
SynSR (known-d)	0.9934	0.9873	0.9813	0.9411	0.9863
SynSR (blind)	0.9879	0.9812	0.9801	0.9375	0.9811

sults suggest that SynSR maintains competitive localization performance across different degradation severities, with more evident benefits under challenging mixed-degradation conditions.

4.7. Ablation study

We perform controlled ablation studies to isolate the contribution of the main components of SynSR. All variants use the same mixed-degradation sampling strategy, training budget, and downstream inspection protocol, with only the target component changed.

As shown in Table 7, introducing the structural prior improves P-AP from 0.5326 to 0.5842 and reduces GradErr from 30.2945 to 28.3642. The Prior + AOR variant further increases P-AP to 0.6127, although PRO decreases slightly from 0.8923 to 0.8862, indicating a minor trade-off between artifact regulation and region-level localization. The Prior + Deg-cond variant achieves a P-AP of 0.6358, a PRO of 0.9237, and a GradErr of 23.1289. The full model achieves the best overall performance, with a P-AP of 0.6659, a PRO of 0.9411, and the lowest GradErr of 20.3258. Relative to the backbone-only variant, the full model improves P-AP by 13.33 percentage points and reduces GradErr by 9.9687, supporting the complementary roles of structural prior guidance, artifact regulation, and degradation conditioning.

We further examine degradation conditioning from the perspectives of learned modulation behavior and blind deployment. Fig. 8 plots the image-wise PriorGate mean against GradErr, with $1 - \mathrm{SSIM}$ used as a degradation proxy. The association quantified by Spearman's ρ shows that the learned gate response varies systematically with degradation-related reconstruction difficulty. Because explicit degradation parameters are generally unavailable in practical deployment, we additionally evaluate a blind variant in which degradation conditioning is disabled by setting $\mathbf{d}_{\mathrm{embed}} = \mathbf{0}$ during both training and testing. As shown in Table 8, the blind variant retains competitive performance, indicating that SynSR can operate effectively without access to known degradation parameters.

Finally, Table 9 examines the contribution of different structural prior sources while keeping the remaining components fixed. The Sobel-only and Laplacian-only variants achieve P-AP values of 0.5327 and 0.5489, respectively, whereas combining the two priors increases P-AP to 0.6659 and reduces GradErr to 20.3258. These results suggest that first-order gradient information and second-order local structural responses provide complementary guidance for restoration. P-AUROC varies only slightly across the three prior configurations, whereas P-AP exhibits a more pronounced difference, indicating that the choice of structural prior has a greater influence on the precision–recall behavior of pixel-level localization in this ablation.

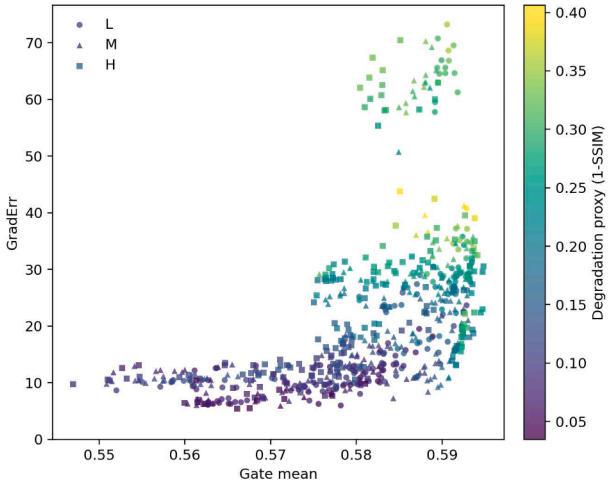


Fig. 8. Gate mean vs. GradErr under mixed degradations at ×4, where each point denotes one test image, point color indicates the degradation proxy $1 - \mathrm{SSIM}$, and Spearman's ρ quantifies the monotonic association.

Table 9
Effect of different structural prior sources. All other components are fixed to the full SynSR model.

Prior source	GradErr↓	P-AUROC↑	P-AP↑
Sobel	21.3681	0.9847	0.5327
Laplacian	22.6871	0.9862	0.5489
Sobel + Laplacian (Ours)	**20.3258**	**0.9873**	**0.6659**

5. Conclusion

In practical production lines, mixed acquisition degradations can weaken subtle defect cues and make inspection more challenging (Hendrycks & Dietterich, 2019). Although super-resolution provides a natural front-end for improving degraded observations, reliable inspection requires not only spatial detail recovery but also control of reconstruction artifacts and spurious high-frequency textures that may interfere with downstream predictions (Molodetskikh et al., 2025).

Motivated by this challenge, we proposed SynSR, an inspection-oriented super-resolution framework for industrial defect inspection under mixed degradations. SynSR integrates structural prior guidance, degradation-aware modulation, and artifact-oriented regulation to improve the reliability of reconstructed information for downstream inspection. Experiments on MVTec AD and MPDD show that SynSR provides strong pixel-level localization performance under the evaluated mixed-degradation conditions, while the ablation studies support the complementary roles of its main components. Nevertheless, the degradations considered in this study are synthetically generated and cannot fully represent device-specific noise, spatially varying blur, illumination and reflection changes, or other production-specific imaging effects. The conclusions should therefore be interpreted within the production-inspired synthetic degradation setting evaluated in this work. Future work will extend SynSR to naturally degraded industrial data and investigate more robust restoration strategies for reducing reconstruction-induced artifacts (Aithal et al., 2024).

CRediT authorship contribution statement

Deyuan Mi: Project administration, Methodology; **Ling Li:** Funding acquisition, Data curation; **Donghao Miao:** Resources, Project administration; **Taoxi Zhu:** Writing – original draft, Validation; **Yaping Jia:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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